

Working with National Crime Victimization Survey Data

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1 Introduction

Many commonly used crime datasets, including the NIBRS data we will encounter in the next set of notes, contain only crimes reported to the police. The National Crime Victimization Survey (NCVS) provides a different perspective by collecting self-reported victimization data. Each year, under the guidance of the Bureau of Justice Statistics, the U.S. Census Bureau interviews a sample of people 12 years old or older about the number and characteristics of crime victimizations they experienced during the prior 6 months.

In 2023, the NCVS completed 226,480 person interviews in 142,028 household interviews. The survey had a 63% response rate for households and an 82% response rate for individuals. Households remain in the sample for $3\frac{1}{2}$ years, completing interviews every 6 months, in person

or by phone, for a total of seven interviews. [Current federal estimates](#) put the annual cost at roughly \$55M and the uncompensated respondent time at roughly 123,000 hours.

The NCVS contains information about nonfatal personal crimes, such as rape and robbery, as well as property crimes, such as burglary. Additional information about the NCVS can be found at the [BJS website](#). To give a sense of the type of data that the NCVS contains, refer to the [Official 2023 BJS Crime Victimization report](#).

2 Acquiring the NCVS data

The University of Michigan consolidates the NCVS data into a format that is easily accessible in R. We will be using data collected in 2022 and 2023 to assemble a dataset that covers victimizations occurring in 2022. Since respondents are asked about crime in the previous six months, respondents completing surveys in May 2023 will still be reporting about crimes in December 2022.

Why am I not using 2024 or 2025 data? In 2024, BJS used a [split-sample design](#), giving half of the randomly selected households the legacy instrument and half the redesigned instrument. BJS has released the legacy data but has not released the data from the half that received the redesigned instrument. The survey fully transitioned to the redesigned instrument in 2025, but BJS has not announced a release date for those data. Data collection for crimes that occurred in 2025 will continue through June 2026. Data collected in 2026 will probably not be posted until fall 2027.

First, we will download the NCVS 2022 data, [ICPSR 38603](#). Click on Download, select R, save the resulting file (called something like ICPSR_38603-V1.zip), extract the contents of the zipped file to a convenient folder, and give it a more understandable folder name, like NCVS2022. Repeat the process for downloading the NCVS 2023 data, [ICPSR 38962](#). New NCVS data tends to appear in mid-September. Typically we need to wait about nine months to get the results from the previous year.

After unzipping the NCVS files, you will find subfolders called DS0001, DS0002, DS0003, DS0004, and DS0005.

```
list.files("NCVS2022/",recursive = TRUE)
```

```
[1] "38603-descriptioncitation.html"
[2] "38603-manifest.txt"
[3] "38603-related_literature.txt"
[4] "38603-User_guide.pdf"
[5] "DS0001/38603-0001-Codebook-ICPSR.pdf"
[6] "DS0001/38603-0001-Data.rda"
[7] "DS0002/38603-0002-Codebook-ICPSR.pdf"
```

```

[8] "DS0002/38603-0002-Data.rda"
[9] "DS0003/38603-0003-Codebook-ICPSR.pdf"
[10] "DS0003/38603-0003-Data.rda"
[11] "DS0004/38603-0004-Codebook-ICPSR.pdf"
[12] "DS0004/38603-0004-Data.rda"
[13] "DS0005/38603-0005-Codebook-ICPSR.pdf"
[14] "DS0005/38603-0005-Data.rda"
[15] "factor_to_numeric_icpsr.R"
[16] "series-95-related_literature.txt"
[17] "TermsOfUse.html"

```

```
list.files("NCVS2023/",recursive = TRUE)
```

```

[1] "38962-descriptioncitation.html"
[2] "38962-manifest.txt"
[3] "38962-related_literature.txt"
[4] "38962-User_guide.pdf"
[5] "DS0001/38962-0001-Codebook-ICPSR.epub"
[6] "DS0001/38962-0001-Codebook-ICPSR.pdf"
[7] "DS0001/38962-0001-Data.rda"
[8] "DS0002/38962-0002-Codebook-ICPSR.epub"
[9] "DS0002/38962-0002-Codebook-ICPSR.pdf"
[10] "DS0002/38962-0002-Data.rda"
[11] "DS0003/38962-0003-Codebook-ICPSR.epub"
[12] "DS0003/38962-0003-Codebook-ICPSR.pdf"
[13] "DS0003/38962-0003-Data.rda"
[14] "DS0004/38962-0004-Codebook-ICPSR.epub"
[15] "DS0004/38962-0004-Codebook-ICPSR.pdf"
[16] "DS0004/38962-0004-Data.rda"
[17] "DS0005/38962-0005-Codebook-ICPSR.epub"
[18] "DS0005/38962-0005-Codebook-ICPSR.pdf"
[19] "DS0005/38962-0005-Data.rda"
[20] "factor_to_numeric_icpsr.R"
[21] "series-95-related_literature.txt"
[22] "TermsOfUse.html"

```

Inside each of these subfolders you will see an R data file with the extension `.rda`. We will spend most of our attention on the contents of the DS0005 folder, which contains the “incident-level extract file.” In each folder you will also find codebooks in PDF (and EPUB) format. The codebook is as important as it is tedious for understanding what is stored in the NCVS data. You should become familiar with the codebooks as soon as you can.

Let's start loading these datasets. We will skip the DS0001 subfolder, which contains basic survey information on the targeted addresses. The DS0002 folder contains data on the households included in the survey.

```
load("NCVS2022/DS0002/38603-0002-Data.rda")
load("NCVS2023/DS0002/38962-0002-Data.rda")

# and let's give them nicer names
dataHH22 <- da38603.0002
dataHH23 <- da38962.0002
```

Take a peek at the first couple of rows.

```
library(dplyr)
library(tidyr)

dataHH22 |>
  head() |>
  select(V2001, YEARQ, IDHH, V2003, V2014, V2016, V2018, V2020,
         V2030, V2031, V2032, V2034, V2036, V2038, V2040A, V2127B, V2129)
```

	V2001	YEARQ	IDHH	V2003
1	(2) Household record	2022.1	1809000258358302568236125 (221)	2022, 1st quarter
2	(2) Household record	2022.1	1809000258380931568236135 (221)	2022, 1st quarter
3	(2) Household record	2022.1	1809000258543680568236125 (221)	2022, 1st quarter
4	(2) Household record	2022.1	1809000284326166568236135 (221)	2022, 1st quarter
5	(2) Household record	2022.1	1809000284384631568236124 (221)	2022, 1st quarter
6	(2) Household record	2022.1	1809000284459399568236114 (221)	2022, 1st quarter

	V2014	V2016	V2018	V2020
1	<NA>	(1) Urban	<NA>	(01) House/apt/flat
2	(1) Owned/being bght	(1) Urban	<NA>	(01) House/apt/flat
3	(1) Owned/being bght	(1) Urban	<NA>	(01) House/apt/flat
4	(1) Owned/being bght	(1) Urban	<NA>	(01) House/apt/flat
5	(1) Owned/being bght	(2) Rural	(2) Less than \$1,000	(01) House/apt/flat
6	(1) Owned/being bght	(2) Rural	(2) Less than \$1,000	(01) House/apt/flat

	V2030	V2031	V2032	V2034
1	(218) Type A - Refused	(01) White only	<NA>	<NA>
2	(300) Interviewed hhld	<NA>	(11) Reference person	(2) Widowed
3	(300) Interviewed hhld	<NA>	(11) Reference person	(3) Divorced
4	(300) Interviewed hhld	<NA>	(11) Reference person	(3) Divorced
5	(300) Interviewed hhld	<NA>	(11) Reference person	(1) Married
6	(300) Interviewed hhld	<NA>	(11) Reference person	(2) Widowed

	V2036	V2038	V2040A	V2127B
1	<NA>	<NA>	<NA>	(3) South
2	(1) Male	(42) Bachelor degree	(01) White only	(3) South
3	(1) Male	(42) Bachelor degree	(01) White only	(3) South
4	(2) Female	(43) Master degree	(01) White only	(3) South
5	(2) Female	(28) High school grad	(01) White only	(3) South
6	(2) Female	(28) High school grad	(01) White only	(3) South

	V2129
1	(2) (S)MSA not city
2	(3) Not (S)MSA
3	(1) City of (S)MSA
4	(2) (S)MSA not city
5	(3) Not (S)MSA
6	(2) (S)MSA not city

The 2022 household dataset has 448 columns. Instead of printing out all of them, I just picked out 17 columns here. First off, you can see that the column names are generally not helpful. That is where the codebook comes in handy. The codebook tells you what each variable means.

Somewhat hidden is a table linking column names to English explanations of what is in those columns. You can get to it by extracting the data frame's "attributes" with `attr()`.

```
varsHH <- dataHH22 |>
  attr("variable.labels") |>
  data.frame() |>
  tibble::rownames_to_column() |>
  setNames(c("varname", "details")) |>
  filter(!grepl("^HHREP", varname)) # exclude rows that start with HHREP
```

Now `varsHH` has two columns, the first with the column names and the second with the details. Let's pull up the 17 columns listed before.

```
varsHH |>
  filter(varname %in% c("V2001", "YEARQ", "IDHH", "V2003", "V2014", "V2016",
    "V2018", "V2020", "V2030", "V2031", "V2032", "V2034", "V2036",
    "V2038", "V2040A", "V2127B", "V2129"))
```

	varname	details
1	V2001	HOUSEHOLD RECORD TYPE
2	YEARQ	YEAR AND QUARTER OF INTERVIEW (YYYY.Q)
3	IDHH	NCVS ID FOR HOUSEHOLDS

4	V2003	YEAR AND QUARTER IDENTIFICATION NUMBER
5	V2014	TENURE (ORIGINAL)
6	V2016	LAND USE (ORIGINAL)
7	V2018	FARM SALES (ORIGINAL)
8	V2020	TYPE OF LIVING QUARTERS (ORIGINAL)
9	V2030	REASON FOR NONINTERVIEW
10	V2031	RACE OF HH HEAD (TYPE A NONINTERVIEW)
11	V2032	PRINCIPAL PERSON RELATION TO REF PERSON
12	V2034	PRINCIPAL PERSON MARITAL STATUS (CURR)
13	V2036	PRINCIPAL PERSON SEX (ALLOCATED)
14	V2038	PRINCIPAL PERSON EDUCATIONAL ATTAINMENT
15	V2040A	PRINCIPAL PERSON RACE RECODE (START 2003 Q1)
16	V2127B	REGION - 1990, 2000, 2010 SAMPLE DESIGN (START 1995 Q3)
17	V2129	MSA STATUS

These are much more intelligible descriptions. “SMSA” stands for Standard Metropolitan Statistical Area, an outdated term. Today we simply call them MSAs. Under the current federal standards, an MSA must contain an urban area with at least 50,000 people.

Note the first household record has IDHH equal to 1809000258543680568236125. We can load the respondent “person file” to see who in this household responded.

```
# loading person-level data
load("NCVS2022/DS0003/38603-0003-Data.rda")
load("NCVS2023/DS0003/38962-0003-Data.rda")
dataPers22 <- da38603.0003
dataPers23 <- da38962.0003

# lookup respondents from this household
dataPers22 |>
  filter(IDHH=="1809000258543680568236125") |>
  select(!starts_with("PERREP")) # drop all the PERREP weight columns
```

	V3001	YEARQ	IDHH	
1	(3) Person record	2022.1	1809000258543680568236125	
2	(3) Person record	2022.3	1809000258543680568236125	
	IDPER	V3002	V3003	V3004
1	180900025854368056823612501	3 (221)	2022, 1st quarter	18
2	180900025854368056823612501	3 (223)	2022, 3rd quarter	18
	V3005	V3006	V3008	V3009
	V3010	V3011		
1	09000258543680568236	1	25	1 (1) Personal/self
2	09000258543680568236	1	25	1 (2) Telephone/self

	V3012	V3013	V3014	V3015	V3016	V3017	V3018
1	(11) Reference person	62	62	(3) Divorced	(3) Divorced	(1) Male	(1) Male
2	(11) Reference person	63	63	(3) Divorced	(3) Divorced	(1) Male	(1) Male
	V3019	V3020	V3023A	V3024	V3024A	V3025	V3026
1	(2) No	(42) Bachelor degree	(01) White only	(2) No	(2) No	(02) February	12
2	(2) No	(42) Bachelor degree	(01) White only	(2) No	(2) No	(08) August	31
	V3027	V3031	V3032	V3033	V3034	V3035	V3040
1	2022	NA	23	NA	(2) No	NA	(2) No
2	2022	NA	24	NA	(2) No	NA	(2) No
	V3046	V3047	V3048	V3049	V3050	V3051	V3052
1	(2) No	NA	(2) No	<NA>	<NA>	<NA>	<NA>
2	(2) No	NA	(2) No	<NA>	<NA>	<NA>	<NA>
	V3058	V3059	V3060	V3061	V3062	V3063	V3064
1	<NA>	NA	(1) At least 1 entry	(0) No	(1) Yes	(0) No	(0) No
2	<NA>	NA	(1) At least 1 entry	(1) Yes	(0) No	(0) No	(0) No
	V3067	V3068	V3069	V3070	V3_V4526H3A	V3_V4526H3B	V3_V4526H5
1	(0) No	(0) No	(0) No	out of range	<NA>	(2) No	(2) No
2	(0) No	(0) No	(0) No	out of range	<NA>	(2) No	(2) No
	V3_V4526H4	V3_V4526H6	V3_V4526H7	V3083			
1	(2) No	(2) No	(2) No	(1) Yes, born in the United States			
2	(2) No	(2) No	(2) No	(1) Yes, born in the United States			
	V3084	V3085	V3086				
1	(2) Straight, that is, not lesbian or gay	(1) Male	(1) Male				
2	(2) Straight, that is, not lesbian or gay	(1) Male	(1) Male				
	V3087	V3088	V3089	V3090	V3091	V3092	V3093
1	(1) Never served in the military	<NA>	<NA>	<NA>	<NA>	<NA>	<NA>
2	(1) Never served in the military	<NA>	<NA>	<NA>	<NA>	<NA>	<NA>
	V3097A	V3098	V3071	V3072	V3073	V3074	V3075
1	<NA>	<NA>	(1) Yes	<NA>	<NA>	(27) Something else	(3) St/cnty/loc govt
2	<NA>	<NA>	(1) Yes	<NA>	<NA>	(27) Something else	(3) St/cnty/loc govt
	V3076	V3077	V3078	V3079	V3080	WGTPERCY	
1	(1) A city	(1) Hhld resp	(2) No	(4) None above schools	1207.963	603.9816	
2	(1) A city	(1) Hhld resp	(2) No	(4) None above schools	1470.257	735.1283	
	V3081	V3082	PER_TIS	PERINTVNUM	PINTTYPE_TIS1	PINTTYPE_TIS2	
1	0	2022	5	5	<NA>	(1) Personal, Self-respondent	
2	0	2022	6	6	<NA>	(1) Personal, Self-respondent	
	PINTTYPE_TIS3	PINTTYPE_TIS4					
1	(2) Telephone, Self-respondent	(2) Telephone, Self-respondent					
2	(2) Telephone, Self-respondent	(2) Telephone, Self-respondent					
	PINTTYPE_TIS5	PINTTYPE_TIS6					
1	(2) Telephone, Self-respondent	(1) Personal, Self-respondent					
2	(2) Telephone, Self-respondent	(1) Personal, Self-respondent					
	PINTTYPE_TIS7	PERBOUNDED					

```

1                                     <NA> (1) Bounded by previous time in sample
2 (2) Telephone, Self-respondent (1) Bounded by previous time in sample

```

These two rows represent two surveys, six months apart, of the same divorced, 62- to 63-year-old white male. Let's look up another household.

```

dataPers22 |>
  filter(IDHH=="1809000284384631568236124") |>
  select(!starts_with("PERREP"))

```

	V3001	YEARQ	IDHH
1	(3) Person record	2022.1	1809000284384631568236124
2	(3) Person record	2022.1	1809000284384631568236124
	IDPER	V3002	V3003 V3004
1	180900028438463156823612401	5 (221) 2022, 1st quarter	18
2	180900028438463156823612402	5 (221) 2022, 1st quarter	18
	V3005 V3006 V3008 V3009 V3010	V3011	
1	09000284384631568236	1 24 1 1 (2) Telephone/self	
2	09000284384631568236	1 24 2 2 (5) Noninterview	
	V3012 V3013 V3014	V3015 V3016 V3017	
1	(11) Reference person	69 69 (1) Married (1) Married (2) Female	
2	(01) Husband	60 60 (1) Married (1) Married (1) Male	
	V3018 V3019	V3020 V3023A V3024 V3024A	
1	(2) Female <NA> (28) High school grad	(01) White only (2) No (2) No	
2	(1) Male (2) No (28) High school grad	(01) White only (2) No (2) No	
	V3025 V3026 V3027 V3031 V3032 V3033 V3034 V3035 V3040 V3041 V3042		
1	(02) February 18 2022 NA 9 NA (2) No NA (2) No NA (2) No		
2	<NA> NA NA NA NA NA <NA> NA <NA> NA <NA>		
	V3043 V3044 V3045 V3046 V3047 V3048 V3049 V3050 V3051 V3052 V3053 V3054		
1	NA (2) No NA (2) No NA (2) No <NA> <NA> <NA> <NA> NA (2) No		
2	NA <NA> NA <NA> NA <NA> <NA> <NA> <NA> <NA> NA <NA>		
	V3055 V3056 V3057 V3058 V3059	V3060 V3061 V3062 V3063	
1	<NA> <NA> <NA> <NA> NA (1) At least 1 entry (1) Yes (0) No (0) No		
2	<NA> <NA> <NA> <NA> NA <NA> <NA> <NA> <NA>		
	V3064 V3065 V3066 V3067 V3068	V3069 V3070 V3_V4526H3A	
1	(0) No (0) No (0) No (0) No (0) No (0) No out of range <NA> (1) Yes		
2	<NA> <NA> <NA> <NA> <NA> <NA> <NA> <NA>		
	V3_V4526H3B V3_V4526H5 V3_V4526H4 V3_V4526H6 V3_V4526H7		
1	(2) No (2) No (2) No (2) No (2) No		
2	<NA> <NA> <NA> <NA> <NA>		
	V3083	V3084	
1	(1) Yes, born in the United States	(2) Straight, that is, not lesbian or gay	

	V3085	V3086		V3087	V3088	V3089	V3090
2			<NA>				<NA>
1	(2) Female	(2) Female	(1) Never served in the military	<NA>	<NA>	<NA>	<NA>
2	<NA>	<NA>		<NA>	<NA>	<NA>	<NA>
	V3091	V3092	V3093	V3094	V3097A	V3098	V3071 V3072 V3073 V3074
1	<NA>	<NA>	<NA>	<NA>	<NA>	<NA>	(1) Yes <NA> (27) Something else
2	<NA>	<NA>	<NA>	<NA>	<NA>	<NA>	<NA>
	V3075	V3076		V3077	V3078		
1	(1) Priv company	(3) Rural area		(1) Hhld resp	(2) No		
2	<NA>	<NA>		(0) Not hhld resp	<NA>		
	V3079	V3080	WGTPERCY	V3081	V3082	PER_TIS	PERINTVNUM
1	(4) None above schools	1070.377	535.1885	0	2022	7	7
2	(4) None above schools	0.000	0.0000	0	2022	7	6
	PINTTYPE_TIS1			PINTTYPE_TIS2			
1	(2) Telephone, Self-respondent	(2) Telephone, Self-respondent					
2	(4) Telephone, Proxy	(4) Telephone, Proxy					
	PINTTYPE_TIS3			PINTTYPE_TIS4			
1	(2) Telephone, Self-respondent	(2) Telephone, Self-respondent					
2	(4) Telephone, Proxy	(4) Telephone, Proxy					
	PINTTYPE_TIS5			PINTTYPE_TIS6			
1	(2) Telephone, Self-respondent	(2) Telephone, Self-respondent					
2	(2) Telephone, Self-respondent	(2) Telephone, Self-respondent					
	PINTTYPE_TIS7			PERBOUNDED			
1	(2) Telephone, Self-respondent	(1) Bounded by previous time in sample					
2	(5) Noninterview	(1) Bounded by previous time in sample					

These rows represent two surveys occurring at the same time. One is of the reference person, a married white female, and the second is of her husband.

Let's grab the variable details as we did with the household data.

```
# Person file also has list of variable details
varsPers <- dataPers22 |>
  attr("variable.labels") |>
  data.frame() |>
  tibble::rownames_to_column() |>
  setNames(c("varname", "details")) |>
  filter(!grepl("^((PERREP|PINTTYPE)", varname))
varsPers
```

	varname
1	V3001

2	YEARQ
3	IDHH
4	IDPER
5	V3002
6	V3003
7	V3004
8	V3005
9	V3006
10	V3008
11	V3009
12	V3010
13	V3011
14	V3012
15	V3013
16	V3014
17	V3015
18	V3016
19	V3017
20	V3018
21	V3019
22	V3020
23	V3023A
24	V3024
25	V3024A
26	V3025
27	V3026
28	V3027
29	V3031
30	V3032
31	V3033
32	V3034
33	V3035
34	V3040
35	V3041
36	V3042
37	V3043
38	V3044
39	V3045
40	V3046
41	V3047
42	V3048
43	V3049
44	V3050

45	V3051
46	V3052
47	V3053
48	V3054
49	V3055
50	V3056
51	V3057
52	V3058
53	V3059
54	V3060
55	V3061
56	V3062
57	V3063
58	V3064
59	V3065
60	V3066
61	V3067
62	V3068
63	V3069
64	V3070
65	V3_V4526H3A
66	V3_V4526H3B
67	V3_V4526H5
68	V3_V4526H4
69	V3_V4526H6
70	V3_V4526H7
71	V3083
72	V3084
73	V3085
74	V3086
75	V3087
76	V3088
77	V3089
78	V3090
79	V3091
80	V3092
81	V3093
82	V3094
83	V3097A
84	V3098
85	V3071
86	V3072
87	V3073

88 V3074
 89 V3075
 90 V3076
 91 V3077
 92 V3078
 93 V3079
 94 V3080
 95 WGTPERCY
 96 V3081
 97 V3082
 98 PER_TIS
 99 PERINTVNUM
 100 PERBOUNDED

1 PERSON RECORD
 2 YEAR AND QUARTER OF INTERVIEW
 3 NCVS ID FOR HOUSEHOLD
 4 NCVS ID FOR PERSON
 5 ICPSR HOUSEHOLD IDENTIFICATION NUMBER
 6 YEAR AND QUARTER IDENTIFICATION
 7 SAMPLE NUMBER
 8 SCRAMBLED CONTROL NUMBER
 9 HOUSEHOLD NUMBER
 10 PANEL AND ROTATION NUMBER
 11 PERSON SEQUENCE NUMBER
 12 PERSON LINE NUMBER
 13 TYPE OF INTERVIEW
 14 RELATIONSHIP TO REFERENCE PERSON
 15 AGE (ORIGINAL)
 16 AGE (ALTERNATE)
 17 MARITAL STATUS (CURRENT)
 18 MARITAL STATUS (PREVIOUS)
 19 SEX (ORIGINAL)
 20 SEX (ALTERNATE)
 21 NOW AN ARMED FORCES MEMBER
 22 EDUCATIONAL ATTAINMENT
 23 RACE RECODE (START 2)
 24 HISPANIC ORIGIN
 25 HISPANIC ORIGIN (ALLOCATED) (START 2)
 26 MONTH INTERVIEW COMPLETED
 27 DAY INTERVIEW COMPLETED
 28 YEAR INTERVIEW COMPLETED
 29 HOW LONG AT THIS ADDRESS

30	HOW LONG AT THIS ADDRESS
31	HOW MANY TIMES MOVED IN LAST
32	SOMETHING STOLEN OR
33	NO. TIMES SOMETHING STOLEN OR
34	ATTACK, THREAT, THEFT: LOCAT
35	NO. TIMES ATTACK, LOCAT
36	ATTACK, THREAT: WEAPON & ATT
37	NO. TIMES ATTACK, WEAP
38	STOLEN, ATTACK, THREAT: OFFENDI
39	NO. TIMES ATTACK, OFFENDI
40	FORCED OR COERCED UNWAI
41	NO. TIMES UNWAI
42	CALL POLICE TO REPORT SOMETH
43	FIRST
44	SECOND
45	THIRD
46	CHECK B: ATTACK, THREAT
47	NO. TIMES ATTACK, THREAT
48	THOUGHT CRIME BUT DIDN'T CAL
49	FIRST
50	SECOND
51	THIRD
52	CHECK C: ATTACK, THREAT
53	NO. TIMES ATTACK, THREAT
54	LI WHO PRESENT DURING SCREEN Q
55	C TELEPHONE IN
56	C NO ONE BESIDES RESPONDENT
57	C RESPONDENT'S
58	C HH MEMBER(S) 12+, NO
59	C HH MEMBER(S) U
60	C NONHOUSEHOLD M
61	C SOMEONE PRESENT, CAN'T
62	C DON'T KNOW IF SOMEONE ELSE
63	RESIDUE: WHO PRESENT DURING
64	DID SELECTED RESPONDENT HEL
65	ARE YOU DEAF OR DO YOU HAVE SERIOUS DIFFICULTY HEARING? (START 2
66	ARE YOU BLIND OR DO YOU HAVE SERIOUS DIFFICULTY SEEING EVEN WHEN WEARING GLASSES (START 2
67	DIFFICULT: LEARN, REMEMBER, CONCENTRATE (START 2
68	LIMITS PHYSICAL ACTIVITIES (START 2
69	DIFFICULT: DRESSING, BATHING, GET AROUND HOME (START 2
70	DIFFICULT: GO OUTSIDE HOME TO SHOP OR DR OFFICE (START 2
71	CITIZENSHIP STATUS (START 2
72	SEXUAL ORIENTATION (START 2

73	GENDER IDENTITY AT BIRTH (START 2
74	CURRENT GENDER IDENTITY (START 2
75	SERVE ON ACTIVE DUTY (START 2
76	LI: WHEN ON ACTIVE DUTY (START 2
77	C ACTIVE DUTY: SEPTEMBER 2001 (START 2
78	C ACTIVE DUTY: AUGUST 1990 TO AUGUST 2001 (START 2
79	C ACTIVE DUTY: MAY 1975 TO JULY 1990 (START 2
80	C ACTIVE DUTY: VIETNAM ERA (AUGUST 1964 TO APRIL 1975) (START 2
81	C ACTIVE DUTY: FEBRUARY 1955 TO JULY 1964 (START 2
82	C ACTIVE DUTY: KOREAN WAR (JULY 1950 TO JANUARY 1955) (START 2
83	C ACTIVE DUTY: DISCLOSURE RECODE (START 2
84	RESIDUE: ACTIVE DUTY (START 2
85	HAVE JOB OR WORK L
86	HAVE JOB OR WORK IN LAST 6
87	DID JOB/WORK LAST 2 WEEKS
88	WHICH BEST DESCRIBES Y
89	IS EMPLOYMENT PRIVATE, GOVT
90	IS WORK MOSTLY IN CITY, SUBUR
91	HOUSEHOLD RE
92	EMPLOYED BY A COLLEGE OR UN
93	ATTENDING
94	PERSON
95	ADJUSTED PERSON WEIGHT - COLLECT
96	NUMBER OF CRIME INCIDENT
97	YEAR IDENTIFICATION (START 2
98	PERSON TIME IN SAMPLE (START 2
99	PERSON INTERVIEW NUMBER (START 2
100	PERSON BOUNDED BY PREVIOUS TIME IN SAMPLE (START 2

There is an incident-level file that we will read in here. We are not going to look at it further, since much of the information in this file is also in the incident extract file.

```
load("NCVS2022/DS0004/38603-0004-Data.rda")
load("NCVS2023/DS0004/38962-0004-Data.rda")
dataInc22 <- da38603.0004
dataInc23 <- da38962.0004

dataInc22 |>
  select(IDHH, IDPER, V4014, V4529) |>
  head()
```

IDHH

IDPER

V4014

```

1 1809010265731899564536114 180901026573189956453611401 (12) December
2 1809040225522254568236115 180904022552225456823611501 (11) November
3 1809213903398449563644234 180921390339844956364423401 (09) September
4 1809240299163750563236135 180924029916375056323613501 (09) September
5 1809243565469154563238115 180924356546915456323811501 (12) December
6 1809243958169129563244125 180924395816912956324412501 (08) August
      V4529
1 (31) Burg, force ent
2 (56) Theft $50-$249
3 (56) Theft $50-$249
4 (56) Theft $50-$249
5 (58) Theft value NA
6 (20) Verbal thr aslt

```

Finally, we will load in the incident extract file and its associated variable details. This extract file merges in household-level and person-level information to the incident-level file, allowing you to connect person-level features with features of the victimizations they report.

```

# incident-level extract file
load("NCVS2022/DS0005/38603-0005-Data.rda")
load("NCVS2023/DS0005/38962-0005-Data.rda")
dataExt22 <- da38603.0005
dataExt23 <- da38962.0005

varsExt <- dataExt22 |>
  attr("variable.labels") |>
  data.frame() |>
  tibble::rownames_to_column() |>
  setNames(c("varname", "details")) |>
  filter(!grepl("INCREPWGT|VICREPWGT", varname))

```

Let's take a look at a few of the reported crime victimizations. Here, I will just pull the respondent's age, marital status, sex, general location, and crime type.

```

dataExt22 |>
  select(V3014, V3015, V3018, V4022, V4529) |>
  slice(1:3)

```

	V3014	V3015	V3018	V4022	V4529
1	56 (3) Divorced	(2) Female	(3) Same city etc	(31) Burg, force ent	
2	78 (2) Widowed	(2) Female	(3) Same city etc	(56) Theft \$50-\$249	
3	43 (1) Married	(1) Male	(3) Same city etc	(56) Theft \$50-\$249	

Not all information from the household and person files is in the extract file, but many of the features that are likely to be of interest are there.

Now that the datasets are loaded and renamed, we can remove objects from our working environment that we no longer need. We can use `rm()` to accomplish this.

```
rm(da38603.0002,da38603.0003,da38603.0004,da38603.0005,  
   da38962.0002,da38962.0003,da38962.0004,da38962.0005)
```

3 Combining 2022 and 2023 data

Here we are going to create a data frame containing all the reported incidents that *occurred* in 2022. Take a look at the month and year of the reported crime incidents.

```
dataExt22 |> count(V4015, V4014)
```

	V4015	V4014	n
1	2021	(07) July	138
2	2021	(08) August	280
3	2021	(09) September	384
4	2021	(10) October	510
5	2021	(11) November	650
6	2021	(12) December	849
7	2022	(01) January	772
8	2022	(02) February	727
9	2022	(03) March	774
10	2022	(04) April	730
11	2022	(05) May	770
12	2022	(06) June	839
13	2022	(07) July	718
14	2022	(08) August	593
15	2022	(09) September	383
16	2022	(10) October	292
17	2022	(11) November	136

```
dataExt23 |> count(V4015, V4014)
```

	V4015	V4014	n
1	2022	(07) July	175
2	2022	(08) August	253

3	2022	(09)	September	364
4	2022	(10)	October	524
5	2022	(11)	November	674
6	2022	(12)	December	859
7	2023	(01)	January	753
8	2023	(02)	February	681
9	2023	(03)	March	699
10	2023	(04)	April	688
11	2023	(05)	May	738
12	2023	(06)	June	760
13	2023	(07)	July	721
14	2023	(08)	August	573
15	2023	(09)	September	447
16	2023	(10)	October	273
17	2023	(11)	November	142

Note that the 2022 NCVS reports on crimes that occurred in 2022 and 2021. Similarly, the 2023 NCVS reports on crimes that occurred in 2023 and 2022. Remember that the NCVS surveys respondents about victimizations from the prior 6 months, excluding the month of the interview. We are going to stack the 2022 and 2023 incident extract data frames and then filter them to exclude 2021 and 2023.

`bind_rows()` stacks data frames on top of each other, useful when combining two datasets that have the same structure. First we will check that they have the same columns in them.

```
identical(names(dataExt22), names(dataExt23))
```

```
[1] TRUE
```

Good so far! Now let's try to stack them.

```
dataExt <- dataExt22 |>
  bind_rows(dataExt23)
```

```
Error in `bind_rows()`:
! Can't combine `..1$V2061` <factor<f6015>> and `..2$V2061` <double>.
```

Hmmm... R is complaining about V2061. Note that it specifically complains that one data frame has V2061 stored as a factor (a categorical variable) and the other one has it stored as a double, a decimal number.

```
dataExt22 |> count(V2061)
```

	V2061	n
1	(01) 1	8
2	<NA>	9537

```
dataExt23 |> count(V2061)
```

	V2061	n
1	1	14
2	3	1
3	4	3
4	NA	9306

What is V2061 anyway?

```
varsExt |> filter(varname=="V2061")
```

	varname	details
1	V2061	LINE NO. OF 4TH PROXY RESPONDENT

This reports on who answered on behalf of an unavailable respondent. It is not really important for us, so let's drop this one by using `select(-V2061)` on both data frames.

```
dataExt <- dataExt22 |>
  select(-V2061) |>
  bind_rows(dataExt23 |>
    select(-V2061))
```

```
Error in `bind_rows()` :
! Can't combine `..1$V4126` <factor<fb04b>> and `..2$V4126` <double>.
```

Ughh. Now it is complaining about V4126.

```
varsExt |> filter(varname=="V4126")
```

	varname	details
1	V4126	WHICH INJURY FROM OTHER WEAPON (3RD)

```
dataExt22 |> count(V4126)
```

	V4126	n
1 (10)	Bruises, cuts	3
2	<NA>	9542

```
dataExt23 |> count(V4126)
```

	V4126	n
1	NA	9324

In the codebook we can find the full question: “Q.33.3 Which injuries were caused by a weapon OTHER than a gun or knife?”. This seems like a potentially interesting question that I probably do not want to discard. The issue is that no 2023 respondent said there was a third weapon that injured them. In 2022 V4126 was stored as a factor, but in 2023, since they are all missing, R defaulted to numeric (double). We can fix this by just telling R to convert the 2023 data into a factor.

```
dataExt <- dataExt22 |>
  select(-V2061) |>
  bind_rows(dataExt23 |>
    select(-V2061) |>
    mutate(V4126=as.factor(V4126)))
```

```
Error in `bind_rows()` :
! Can't combine `..1$V4313` <factor<514cc>> and `..2$V4313` <double>.
```

Dammit! Now it is complaining about V4313. What is the problem with this one? Again we have a problem with 2022 storing as a factor and 2023 storing as a double.

```
varsExt |> filter(varname=="V4313")
```

	varname	details
1	V4313	3RD LINE NO. OF OTHER OWNER THEFT ITEMS

```
dataExt22 |> count(V4313)
```

	V4313	n
1 (06)	6	2
2	<NA>	9543

```
dataExt23 |> count(V4313)
```

	V4313	n
1	4	1
2	5	1
3	NA	9322

This column answers “Besides the respondent, which household member(s) owned the (property/money) the offender tried to take?” In 2022, the only responses were missing or #6. In 2023, the responses were missing, 4, or 5. These should be numbers since they are supposed to link to respondents in the same household affected by the theft. The approach I’ll take is to use `recode_values()`, telling R to change the 2022 “(06) 6” response to a regular 6.

```
dataExt <- dataExt22 |>
  select(-V2061) |>
  mutate(V4313 = recode_values(V4313,
                              "(06) 6" ~ 6,
                              default = NA)) |>

  bind_rows(dataExt23 |>
    select(-V2061) |>
    mutate(V4126=as.factor(V4126)))
```

Error in `bind\_rows()`:

! Can't combine `..1\$V4357A` <double> and `..2\$V4357A` <factor<758b7>>.

Will this ever stop?!?!? Another double in one year and factor in another year, this time affecting V4357A asking about handguns.

```
varsExt |> filter(varname=="V4357A")
```

	varname	details
1	V4357A	HOW MANY HANDGUNS WERE TAKEN (START 2004 Q1)

```
dataExt22 |> count(V4357A)
```

	V4357A	n
1	1	41
2	2	8

3	3	3
4	997	1
5	998	1
6	NA	9491

```
dataExt23 |> count(V4357A)
```

	V4357A	n
1	(001) 1	48
2	(998) Residue	1
3	<NA>	9275

You might have seen this word “Residue” show up before. For the NCVS, BJS records “Residue” when there is a data entry error resulting in an out-of-range code, an incorrect or unusable answer by the respondent, or the absence of an entry for a question that should have been asked. Sometimes you might also see “Out of universe/blank.” This happens when a question is outside the set that a respondent should answer. For example, only victims who report being injured are asked whether they received medical care. All other victims skip the question.

I will solve this issue by recoding the 2023 values to numeric values.

```
dataExt <- dataExt22 |>
  select(-V2061) |>
  mutate(V4313 = recode_values(V4313,
                              "(06) 6" ~ 6,
                              default = NA)) |>
  bind_rows(dataExt23 |>
    select(-V2061) |>
    mutate(V4126 = as.factor(V4126),
           V4357A = recode_values(V4357A,
                                   "(001) 1" ~ 1,
                                   "(998) Residue" ~ 998,
                                   default=NA)))
```

Success! Now let’s check that all is okay now.

```
dataExt |> count(V4126)
```

	V4126	n
1	(10) Bruises, cuts	3
2	<NA>	18866

```
dataExt |> count(V4313)
```

	V4313	n
1	4	1
2	5	1
3	6	2
4	NA	18865

```
dataExt |> count(V4357A)
```

	V4357A	n
1	1	89
2	2	8
3	3	3
4	997	1
5	998	2
6	NA	18766

Remember that we still have data in here from 2021 and 2023.

```
table(dataExt$V4015)
```

2021	2022	2023
2811	9583	6475

We are just going to focus on 2022.

```
dataExt <- dataExt |> filter(V4015==2022)
```

Note that BJS official reports generally classify victimizations by the year of the survey and not by the year of the crime. Our 2022 calendar-year dataset combines records carrying collection-year weights from two survey years, so its estimates will not exactly reproduce the official 2022 estimates.

Some respondents report crime victimizations that occurred outside of the United States.

```
# V4022 - IN WHAT CITY, TOWN, VILLAGE.  
dataExt |> count(V4022)
```

	V4022	n
1	(1) Outside U.S.	44
2	(2) Not in city etc	67
3	(3) Same city etc	8069
4	(4) Diff city etc	1354
5	(5) Don't know	37
6	(8) Residue	12

The BJS convention is to exclude these crimes in official reports (see 2023 User Guide, page 21).

```
dataExt <- dataExt |>
  filter(is.na(V4022) | V4022!="(1) Outside U.S.")
```

4 Sampling weights

A sampling weight tells us how many units in the population a sampled record represents. The NCVS interviews a sample of U.S. households and people, so one record in the data may represent hundreds or thousands of victimizations in the population. To estimate a national total, we sum the weights rather than count the rows. To estimate a percentage, we sum the weights for records with the feature of interest and divide by the sum of all the relevant weights. The weights account for the sample design, adjust for nonresponse, and calibrate the sample to known population totals.

The NCVS sampling design has historically emphasized stability and operational continuity, relying heavily on Census Bureau sampling and fieldwork logistics. A [2008 National Academies report](#) noted that the original design gave each household the same probability of selection, even though such a design was not optimal for many uses of the NCVS, and recommended giving housing units higher selection probabilities where victimization rates were higher. At the first stage, the design has generally sampled counties with probabilities approximately proportional to population size, without including a local crime-rate measure. Research suggests that police-reported crime rates and tract-level demographic measures can predict victimization and could improve precision, but those measures are not currently part of the sampling plan. The sampling weights are still essential because selection probabilities vary for other design reasons, and the weights also adjust for nonresponse and calibrate the sample to known population totals.

Constructing the sampling weights is a complex process (see the User Guide “Weights Details” section starting on page 22). The NCVS sampling weights adjust for six factors. From the User Guide:

1. *Base weight*: The inverse of the national sampling rate for the stratum of that unit (person or household)
2. *Weighting control*: Adjusts for any subsampling due to unexpected events in the field, such as new construction, area segments larger than anticipated, and other deviations from the overall stratum sampling rate
3. *Household noninterview adjustment*: Adjusts for nonresponse at the household level by increasing the weights of interviewed households most similar to households not interviewed in terms of race, MSA status of residence, and urban/suburban/rural status of residence. This inflates the weight assigned to interviewed households so that they represent themselves and noninterviewed households. The noninterviewed cases are assigned a weight of zero, excluding them from population estimates
4. *Within-household noninterview adjustment*: Adjusts for nonresponse at the person level by increasing the weights of interviewed persons most similar to persons not interviewed in terms of region, age, race, sex, and household composition. The adjustment inflates the weight assigned to completed interviews so that they represent themselves and sampled individuals who were not interviewed. The noninterviewed cases are assigned a weight of zero
5. *First-stage ratio estimate factor*: Adjusts for differences between the characteristics of sampled non-self-representing (NSR) primary sampling units (PSUs) and independent measures of the population in NSR PSUs. For self-representing PSUs, this factor is set to 1. The factor adjusts for PSU differences in region, MSA status, urban/suburban/rural status, and racial composition. (PSUs are essentially counties... or a small group of counties.)
6. *Second-stage ratio estimate factor*: A poststratification factor defined for each person to adjust for the difference between weighted counts of persons, using the five weight components above, and independent estimates of the number of persons within certain age-by-race-by-sex categories. These independent estimates are based on Census population controls adjusted for the undercount

 Always use the sampling weights

Importantly, every population estimate you produce with the NCVS must use the appropriate weights. Use **SERIES_WEIGHT** for victimization totals and percentages, and use **SERIES_IWEIGHT** when the unit of interest is a criminal incident. To estimate a victimization rate, divide the weighted number of victimizations by the weighted population at risk. Use person weights for personal crimes (**WGTPERCY**) and household weights (**WGTHHCY**) for household property crimes. Even plots and figures intended to describe the U.S. population should use weights. Unweighted tabulations are still useful for checking the data, but they should not be presented as national estimates.

Where you would normally compute a sample mean as $\frac{\sum x_i}{n}$, the weighted mean is

$$\frac{\sum w_i x_i}{\sum w_i}$$

For a weighted percentage, total all the weights for records with the particular feature and divide by the total weight. For some plots there is not an obvious way to accommodate the sampling weights. In those cases, we can sample with replacement with probabilities proportional to the sampling weights and plot the sampled points.

This chapter focuses on point estimates. To calculate valid standard errors and confidence intervals, you also need to account for the complex sample design. The incident extract includes 160 replicate weights for this purpose, and BJS uses balanced repeated replication with Fay’s method.

For estimating victimization totals in the incident extract, we will focus on two weights. **WGTVICCY** is the adjusted collection-year victimization weight. It tells us how many victimizations a record represents when that record is counted once. **SERIES_WEIGHT** begins with **WGTVICCY** and also accounts for series crimes.

Some crimes happen in a series. For example, a respondent may report regular domestic abuse that happened numerous times over the last six months. Each occurrence of domestic abuse is a victimization, but the BJS convention is to count up to 10 occurrences for crimes reported as a series (2023 User Guide, pages 18-19). For a nonseries crime, **SERIES_WEIGHT** and **WGTVICCY** are equal. For a series crime, **SERIES_WEIGHT** is **WGTVICCY** multiplied by the reported number of occurrences, capped at 10. If the number of occurrences is unavailable, BJS uses six.

Variable **V4016** records the answer to “Altogether, how many times did this type of incident happen during the last 6 months?” and variable **V4019** documents “Can you (respondent) recall enough details of each incident to distinguish them from each other?” Note that the coding of **V4016** has 997 representing “Don’t know” and 998 representing “Residue.” These are not counts of victimizations. Fortunately, **SERIES_WEIGHT** already incorporates the series adjustment, so we do not need to change **V4016** ourselves.

The ratio of **SERIES_WEIGHT** to **WGTVICCY** shows how many victimizations each record contributes after the series adjustment. The ratio is 1 for nonseries crimes and ranges from 2 to 10 for the series crimes shown below.

```
dataExt |>
  mutate(seriesMultiplier = SERIES_WEIGHT/WGTVICCY) |>
  filter(seriesMultiplier > 1) |>
  select(V4016, V4019, WGTVICCY, SERIES_WEIGHT, seriesMultiplier) |>
  head()
```

V4016	V4019	WGTVICCY	SERIES_WEIGHT	seriesMultiplier
-------	-------	----------	---------------	------------------

1	7	(2)	No (is series)	2315.6980	16209.886	7
2	6	(2)	No (is series)	418.0296	2508.178	6
3	6	(2)	No (is series)	1899.7567	11398.540	6
4	6	(2)	No (is series)	2235.6122	13413.673	6
5	7	(2)	No (is series)	1560.4005	10922.804	7
6	30	(2)	No (is series)	3059.0421	30590.421	10

Exercises

1. Compare `nrow(dataExt)`, `sum(dataExt$WGTVICCY)`, and `sum(dataExt$SERIES_WEIGHT)`. What does each number represent?
2. Calculate the percentage increase from `WGTVICCY` to `SERIES_WEIGHT`. This is the effect of accounting for series victimizations on the estimated total.

5 Tabulating victimizations

First, we need to be clear about what we are counting. BJS reports on *victimizations* and *incidents*. For personal crimes, victimizations count the number of people victimized. For household property crimes, the affected household is treated as the victim. Incidents count criminal events, and a single incident can involve multiple victims. BJS reports largely focus on criminal victimizations.

We can start by just asking the NCVS data how many criminal victimizations there were in 2022. We compute that as the sum of all the weights.

```
sum(dataExt$SERIES_WEIGHT)
```

```
[1] 20121320
```

This means that the NCVS estimates that there were 20,121,320 criminal victimizations in the United States in 2022.

Let's take a closer look at what kinds of victimization occurred. Note that this code breaks the dataset into groups based on the reported crime type, `V4529`, and computes the total weight associated with each of those crime categories.

```
dataExt |>
  group_by(V4529) |> # crime type
  summarize(total = sum(SERIES_WEIGHT)) |>
  print(n=Inf)
```

A tibble: 33 x 2

V4529	total
<fct>	<dbl>
1 (01) Completed rape	127522.
2 (02) Attempted rape	67330.
3 (03) Sex aslt w s aslt	27038.
4 (04) Sex aslt w m aslt	7511.
5 (05) Rob w inj s aslt	87026.
6 (06) Rob w inj m aslt	88620.
7 (07) Rob wo injury	216589.
8 (08) At rob inj s asl	25197.
9 (09) At rob inj m asl	47896.
10 (10) At rob w aslt	119279.
11 (11) Ag aslt w injury	449842.
12 (12) At ag aslt w wea	397249.
13 (13) Thr aslt w weap	637952.
14 (14) Simp aslt w inj	578068.
15 (15) Sex aslt wo inj	118078.
16 (16) Unw sex wo force	52425.
17 (17) Asl wo weap, wo inj	1295757.
18 (18) Verbal thr rape	41862.
19 (19) Ver thr sex aslt	15501.
20 (20) Verbal thr aslt	1978623.
21 (21) Purse snatching	35375.
22 (23) Pocket picking	124503.
23 (31) Burg, force ent	549203.
24 (32) Burg, ent wo for	1074214.
25 (33) Att force entry	312530.
26 (40) Motor veh theft	518797.
27 (41) At mtr veh theft	193058.
28 (54) Theft < \$10	623035.
29 (55) Theft \$10-\$49	1735619.
30 (56) Theft \$50-\$249	2968819.
31 (57) Theft \$250+	3057945.
32 (58) Theft value NA	1799273.
33 (59) Attempted theft	749585.

The first 20 crime types listed are violent crimes, codes 21 through 23 are personal larceny (like pickpocketing... no violence or breaking into a home), and the remainder are household property crimes. For household property crimes, the NCVS treats the household, rather than an individual respondent, as the victim. Let's extract the two-digit code between the parentheses so that we can classify the crime types.

```
dataExt <- dataExt |>
  mutate(crimeCode = as.integer(gsub("^\\((([0-9]{2})\\)).*", "\\1", V4529)),
         crimeType = case_when(
           crimeCode <= 20 ~ "violent",
           crimeCode <= 23 ~ "personal larceny",
           .default = "property"))

dataExt |>
  select(V4529, crimeCode, crimeType) |>
  head()
```

	V4529	crimeCode	crimeType
1	(41) At mtr veh theft	41	property
2	(59) Attempted theft	59	property
3	(58) Theft value NA	58	property
4	(57) Theft \$250+	57	property
5	(55) Theft \$10-\$49	55	property
6	(07) Rob wo injury	7	violent

That all looks correct, so now I can move on to the tabulation.

```
a <- dataExt |>
  group_by(crimeType) |>
  summarize(estTotal = sum(SERIES_WEIGHT))
a
```

```
# A tibble: 3 x 2
  crimeType      estTotal
  <chr>         <dbl>
1 personal larceny 159878.
2 property       13582078.
3 violent        6379364.
```

In our calendar-year 2022 dataset, there were an estimated 6,379,364 violent victimizations, 159,878 personal larcenies, and 13,582,078 household property victimizations.

We can summarize other categories, like car thefts, attempted and completed.

```
dataExt |>
  filter(V4529 %in% c("(40) Motor veh theft",
                     "(41) At mtr veh theft")) |>
  summarize(sum(SERIES_WEIGHT))
```

```
sum(SERIES_WEIGHT)
1          711855.4
```

Measuring sexual assault has been complicated by numerous, sometimes major, changes (improvements, more precisely) in the definitions and data collection methods (e.g., question wording). The FBI's Uniform Crime Reporting Program made a major change to the [definition of rape](#) in 2013. The NCVS began phasing in its most recent instrument change in 2024. See Fisher and Gross (2025) for an extended discussion and timeline of the changes.

Here is the NCVS estimate of the number of rapes (attempted and completed) for 2022.

```
dataExt |>
  filter(V4529 %in% c("(01) Completed rape",
                     "(02) Attempted rape")) |>
  summarize(sum(SERIES_WEIGHT))
```

```
sum(SERIES_WEIGHT)
1          194852.7
```

Official NCVS reports combine rape and sexual assault. There are a lot of crime categories that describe these victimizations.

```
unique(dataExt$V4529)
```

```
[1] (41) At mtr veh theft      (59) Attempted theft      (58) Theft value NA
[4] (57) Theft $250+          (55) Theft $10-$49        (07) Rob wo injury
[7] (13) Thr aslt w weap      (17) Asl wo weap, wo inj (20) Verbal thr aslt
[10] (32) Burg, ent wo for     (56) Theft $50-$249      (14) Simp aslt w inj
[13] (54) Theft < $10         (31) Burg, force ent     (19) Ver thr sex aslt
[16] (23) Pocket picking       (10) At rob w aslt       (11) Ag aslt w injury
[19] (40) Motor veh theft      (33) Att force entry     (12) At ag aslt w wea
[22] (04) Sex aslt w m aslt    (21) Purse snatching     (03) Sex aslt w s aslt
[25] (16) Unw sex wo force     (09) At rob inj m asl    (02) Attempted rape
[28] (06) Rob w inj m aslt     (15) Sex aslt wo inj     (18) Verbal thr rape
[31] (01) Completed rape       (08) At rob inj s asl    (05) Rob w inj s aslt
34 Levels: (01) Completed rape (02) Attempted rape ... (59) Attempted theft
```

Let's count any that have the word "rape" and any that have the word "sex" (sometimes capitalized). Here are the ones that BJS counts in this category.

```
dataExt |>
  filter(grepl("rape|[Ss]ex", V4529)) |>
  distinct(V4529)
```

```

      V4529
1  (19) Ver thr sex aslt
2 (04) Sex aslt w m aslt
3 (03) Sex aslt w s aslt
4  (16) Unw sex wo force
5   (02) Attempted rape
6  (15) Sex aslt wo inj
7  (18) Verbal thr rape
8   (01) Completed rape
```

The estimated number of all sexual assaults in 2022 in the United States is

```
dataExt |>
  filter(grepl("rape|[Ss]ex", V4529)) |>
  summarize(sum(SERIES_WEIGHT))
```

```

      sum(SERIES_WEIGHT)
1                457268
```

Since 2020, we have not had national police-reported crime estimates. The FBI transitioned away from the traditional Summary Reporting System (SRS) series in 2021, and its replacement, NIBRS, has had low participation rates with minimal reporting from California, New York, Florida, and Pennsylvania.

The NCVS's value lies in its inclusion of victimizations that were never reported to police.

Exercises

3. Using `crimeType`, calculate the weighted percentage of all victimizations in each of the three categories: violent crime, personal larceny, and household property crime
4. Estimate the number and weighted percentage of motor vehicle thefts that were attempts rather than completed thefts

6 Calculating victimization by demographics

In the remainder of these notes, we will examine relationships between personal victimization and respondents' features, like age (V3014), marital status (V3015), and sex (V3018). To make the code clearer, let's give these variables more intelligible names.

```
dataExt <- dataExt |>
  rename(age=V3014, marital=V3015, sex=V3018)
```

Perhaps we are interested in which personal crimes disproportionately affect men and which disproportionately affect women. Household property crimes are attributed to the household rather than to an individual victim, so we will exclude them from these demographic comparisons. Start by tabulating personal crime type by sex.

```
dataExt |>
  filter(crimeCode <= 23) |>
  group_by(V4529, sex) |>
  summarize(count=sum(SERIES_WEIGHT)) |>
  print(n=Inf)
```

```
`summarise()` has regrouped the output.
i Summaries were computed grouped by V4529 and sex.
i Output is grouped by V4529.
i Use `summarise(.groups = "drop_last")` to silence this message.
i Use `summarise(.by = c(V4529, sex))` for per-operation grouping
  (`?dplyr::dplyr_by`) instead.
```

```
# A tibble: 43 x 3
# Groups:   V4529 [22]
  V4529          sex      count
  <fct>      <fct>      <dbl>
1 (01) Completed rape (1) Male    18080.
2 (01) Completed rape (2) Female 109442.
3 (02) Attempted rape (1) Male     2190.
4 (02) Attempted rape (2) Female  65140.
5 (03) Sex aslt w s aslt (2) Female  27038.
6 (04) Sex aslt w m aslt (1) Male      677.
7 (04) Sex aslt w m aslt (2) Female   6834.
8 (05) Rob w inj s aslt (1) Male   52000.
9 (05) Rob w inj s aslt (2) Female  35026.
10 (06) Rob w inj m aslt (1) Male   16581.
```

11	(06)	Rob w inj m aslt	(2)	Female	72039.
12	(07)	Rob wo injury	(1)	Male	102379.
13	(07)	Rob wo injury	(2)	Female	114209.
14	(08)	At rob inj s asl	(1)	Male	16131.
15	(08)	At rob inj s asl	(2)	Female	9065.
16	(09)	At rob inj m asl	(1)	Male	31641.
17	(09)	At rob inj m asl	(2)	Female	16255.
18	(10)	At rob w aslt	(1)	Male	57154.
19	(10)	At rob w aslt	(2)	Female	62125.
20	(11)	Ag aslt w injury	(1)	Male	195301.
21	(11)	Ag aslt w injury	(2)	Female	254541.
22	(12)	At ag aslt w wea	(1)	Male	189557.
23	(12)	At ag aslt w wea	(2)	Female	207692.
24	(13)	Thr aslt w weap	(1)	Male	402951.
25	(13)	Thr aslt w weap	(2)	Female	235001.
26	(14)	Simp aslt w inj	(1)	Male	214975.
27	(14)	Simp aslt w inj	(2)	Female	363093.
28	(15)	Sex aslt wo inj	(1)	Male	22433.
29	(15)	Sex aslt wo inj	(2)	Female	95645.
30	(16)	Unw sex wo force	(1)	Male	7980.
31	(16)	Unw sex wo force	(2)	Female	44446.
32	(17)	Asl wo weap, wo inj	(1)	Male	648373.
33	(17)	Asl wo weap, wo inj	(2)	Female	647384.
34	(18)	Verbal thr rape	(1)	Male	15134.
35	(18)	Verbal thr rape	(2)	Female	26727.
36	(19)	Ver thr sex aslt	(1)	Male	5480.
37	(19)	Ver thr sex aslt	(2)	Female	10021.
38	(20)	Verbal thr aslt	(1)	Male	1021068.
39	(20)	Verbal thr aslt	(2)	Female	957555.
40	(21)	Purse snatching	(1)	Male	1947.
41	(21)	Purse snatching	(2)	Female	33428.
42	(23)	Pocket picking	(1)	Male	92254.
43	(23)	Pocket picking	(2)	Female	32249.

R produces a long narrow table. This format is sometimes useful, particularly when merging data frames. However, in this case having a table with counts for men and women side-by-side would be easier to absorb. `pivot_wider()` will swing the sex column into two side-by-side columns.

```
dataExt |>
  filter(crimeCode <= 23) |>
  group_by(V4529, sex) |>
```



```
summarize(count=sum(SERIES_WEIGHT)) |>
pivot_wider(names_from=sex,
             values_from=count,
             values_fill = 0) |> # fill in NAs with 0
print(n=Inf)
```

`summarise()` has regrouped the output.
 i Summaries were computed grouped by V4529 and sex.
 i Output is grouped by V4529.
 i Use `summarise(.groups = "drop\_last")` to silence this message.
 i Use `summarise(.by = c(V4529, sex))` for per-operation grouping
 (`?dplyr::dplyr\_by`) instead.

```
# A tibble: 22 x 3
# Groups:   V4529 [22]
  V4529                `(1) Male` `(2) Female`
  <fct>                <dbl>      <dbl>
1 (01) Completed rape    18080.    109442.
2 (02) Attempted rape    2190.     65140.
3 (03) Sex aslt w s aslt      0      27038.
4 (04) Sex aslt w m aslt    677.     6834.
5 (05) Rob w inj s aslt   52000.    35026.
6 (06) Rob w inj m aslt   16581.    72039.
7 (07) Rob wo injury    102379.   114209.
8 (08) At rob inj s asl   16131.     9065.
9 (09) At rob inj m asl   31641.    16255.
10 (10) At rob w aslt     57154.    62125.
11 (11) Ag aslt w injury  195301.   254541.
12 (12) At ag aslt w wea  189557.   207692.
13 (13) Thr aslt w weap   402951.   235001.
14 (14) Simp aslt w inj   214975.   363093.
15 (15) Sex aslt wo inj   22433.    95645.
16 (16) Unw sex wo force   7980.    44446.
17 (17) Asl wo weap, wo inj 648373.   647384.
18 (18) Verbal thr rape   15134.    26727.
19 (19) Ver thr sex aslt   5480.    10021.
20 (20) Verbal thr aslt  1021068.   957555.
21 (21) Purse snatching   1947.    33428.
22 (23) Pocket picking   92254.    32249.
```

Lastly, let's normalize the columns so that they add up to 100%, giving us the distribution of crime types within each sex.

```
dataExt |>
  filter(crimeCode <= 23) |>
  group_by(V4529, sex) |>
  summarize(count=sum(SERIES_WEIGHT)) |>
  ungroup() |> # without this, percentages are computed within crime type
  pivot_wider(names_from = sex,
              values_from = count,
              values_fill = 0) |>
  rename(male=`(1) Male`, female=`(2) Female`) |>
  mutate(male = 100*male/sum(male),
         female=100*female/sum(female),
         ratio=female/male) |>
  arrange(desc(ratio)) |>
  print(n=Inf)
```

`summarise()` has regrouped the output.
 i Summaries were computed grouped by V4529 and sex.
 i Output is grouped by V4529.
 i Use `summarise(.groups = "drop\_last")` to silence this message.
 i Use `summarise(.by = c(V4529, sex))` for per-operation grouping
 (`?dplyr::dplyr\_by`) instead.

A tibble: 22 x 4

	V4529	male	female	ratio
	<fct>	<dbl>	<dbl>	<dbl>
1	(03) Sex aslt w s aslt	0	0.789	Inf
2	(02) Attempted rape	0.0703	1.90	27.0
3	(21) Purse snatching	0.0625	0.976	15.6
4	(04) Sex aslt w m aslt	0.0217	0.200	9.18
5	(01) Completed rape	0.581	3.20	5.50
6	(16) Unw sex wo force	0.256	1.30	5.06
7	(06) Rob w inj m aslt	0.532	2.10	3.95
8	(15) Sex aslt wo inj	0.720	2.79	3.88
9	(19) Ver thr sex aslt	0.176	0.293	1.66
10	(18) Verbal thr rape	0.486	0.780	1.61
11	(14) Simp aslt w inj	6.90	10.6	1.54
12	(11) Ag aslt w injury	6.27	7.43	1.19
13	(07) Rob wo injury	3.29	3.33	1.01
14	(12) At ag aslt w wea	6.09	6.06	0.996
15	(10) At rob w aslt	1.84	1.81	0.988
16	(17) Asl wo weap, wo inj	20.8	18.9	0.908

17	(20)	Verbal thr aslt	32.8	28.0	0.853
18	(05)	Rob w inj s aslt	1.67	1.02	0.612
19	(13)	Thr aslt w weap	12.9	6.86	0.530
20	(08)	At rob inj s asl	0.518	0.265	0.511
21	(09)	At rob inj m asl	1.02	0.475	0.467
22	(23)	Pocket picking	2.96	0.942	0.318

Sexual assaults disproportionately affect women, while pocket picking and attempted robbery involving assaults disproportionately affect men.

The sequence `group_by()`, `summarize()`, `ungroup()` is so common that there is an alternative way to do the same calculation more compactly with `.by` in `summarize()`. A frequent R error is to use `group_by()`, then forget that the data is still grouped, and continue to do calculations unaware that they are occurring only within groups. The `.by` argument also helps avoid this error.

```
# can reduce the group_by, summarize, ungroup with .by
dataExt |>
  filter(crimeCode <= 23) |>
  summarize(count=sum(SERIES_WEIGHT),
            .by = c(V4529, sex)) |> # eliminates group/ungroup
  pivot_wider(names_from = sex,
              values_from = count,
              values_fill = 0) |>
  rename(male=`(1) Male`, female=`(2) Female`) |>
  mutate(male = 100*male/sum(male),
         female=100*female/sum(female),
         ratio=female/male) |>
  arrange(desc(ratio)) |>
  print(n=Inf)
```

```
# A tibble: 22 x 4
  V4529          male female  ratio
  <fct>      <dbl>   <dbl>  <dbl>
1 (03) Sex aslt w s aslt    0    0.789 Inf
2 (02) Attempted rape    0.0703  1.90  27.0
3 (21) Purse snatching  0.0625  0.976  15.6
4 (04) Sex aslt w m aslt  0.0217  0.200   9.18
5 (01) Completed rape    0.581   3.20   5.50
6 (16) Unw sex wo force  0.256   1.30   5.06
7 (06) Rob w inj m aslt   0.532   2.10   3.95
8 (15) Sex aslt wo inj    0.720   2.79   3.88
```

9	(19)	Ver thr sex aslt	0.176	0.293	1.66
10	(18)	Verbal thr rape	0.486	0.780	1.61
11	(14)	Simp aslt w inj	6.90	10.6	1.54
12	(11)	Ag aslt w injury	6.27	7.43	1.19
13	(07)	Rob wo injury	3.29	3.33	1.01
14	(12)	At ag aslt w wea	6.09	6.06	0.996
15	(10)	At rob w aslt	1.84	1.81	0.988
16	(17)	Asl wo weap, wo inj	20.8	18.9	0.908
17	(20)	Verbal thr aslt	32.8	28.0	0.853
18	(05)	Rob w inj s aslt	1.67	1.02	0.612
19	(13)	Thr aslt w weap	12.9	6.86	0.530
20	(08)	At rob inj s asl	0.518	0.265	0.511
21	(09)	At rob inj m asl	1.02	0.475	0.467
22	(23)	Pocket picking	2.96	0.942	0.318

We can do a similar calculation by age. First, let's discretize age into some fixed age bins. Then, we can repeat the same calculation to learn about victimization differences by age. I have sorted the results by the 18-24 age category, but you can change it to your age category if you wish.

```
# can reduce the group_by, summarize, ungroup with .by
dataExt |>
  filter(crimeCode <= 23) |>
  mutate(ageGroup =
    cut(age,
      breaks = c(0, 17, 24, 34, 49, 64, Inf),
      labels = c("12-17", "18-24", "25-34",
        "35-49", "50-64", "65+"))) |>
  summarize(count=sum(SERIES_WEIGHT),
    .by = c(V4529, ageGroup)) |>
  pivot_wider(names_from = ageGroup,
    values_from = count,
    values_fill = 0,
    names_sort = TRUE) |> # keep age groups ordered
  # apply the same function to every column, except V4529
  mutate(across(-V4529, function(x) 100*x/sum(x))) |>
  arrange(desc(`18-24`)) |> # you can change to your age group
  print(n=Inf)
```

A tibble: 22 x 7

V4529	`12-17`	`18-24`	`25-34`	`35-49`	`50-64`	`65+`
<fct>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>

1	(20)	Verbal thr aslt	20.1	23.3	31.6	29.9	37.7	35.6
2	(17)	Asl wo weap, wo inj	33.2	14.9	19.6	24.2	13.2	18.3
3	(13)	Thr aslt w weap	5.12	11.3	10.4	10.8	10.3	6.71
4	(12)	At ag aslt w wea	6.91	8.45	4.29	6.61	5.89	4.12
5	(11)	Ag aslt w injury	4.79	8.35	7.15	7.05	7.94	2.89
6	(14)	Simp aslt w inj	12.9	8.13	10.1	7.28	8.60	7.11
7	(15)	Sex aslt wo inj	3.24	4.18	1.35	0.944	1.39	0
8	(01)	Completed rape	0.284	3.80	3.23	1.78	0.206	1.52
9	(07)	Rob wo injury	2.50	3.24	3.09	2.65	3.66	5.84
10	(06)	Rob w inj m aslt	1.34	2.99	2.74	0.0720	0.152	0.877
11	(02)	Attempted rape	0.338	2.61	0.352	1.30	0.729	0.489
12	(21)	Purse snatching	0	2.47	0	0	0.405	0.563
13	(10)	At rob w aslt	0.840	1.53	1.59	1.58	2.56	3.04
14	(23)	Pocket picking	2.34	1.03	1.37	1.69	1.85	5.11
15	(08)	At rob inj s asl	1.38	0.943	0.255	0.155	0	0
16	(16)	Unw sex wo force	3.06	0.919	0.375	0	0.837	1.09
17	(18)	Verbal thr rape	0.181	0.719	0.491	0.640	1.18	0.143
18	(05)	Rob w inj s aslt	0	0.443	1.24	0.891	2.88	2.34
19	(03)	Sex aslt w s aslt	0	0.408	0.200	0	0	3.44
20	(09)	At rob inj m asl	0.512	0.315	0.374	1.96	0.312	0.459
21	(19)	Ver thr sex aslt	0.414	0	0.360	0.317	0.0569	0.382
22	(04)	Sex aslt w m aslt	0.590	0	0	0.145	0.117	0

Exercise

- Repeat the demographic analysis using marital status. For each marital-status group, calculate the weighted percentage of personal victimizations in every crime category. Which crime types make up a larger percentage among divorced respondents than among married respondents?

Since we have made many changes to the dataset, I find it useful to save the final version. This way I can simply `load()` the data again later and know that it already has all the edits and changes that I have made.

```
save(dataExt, file="NCVS2022.RData", compress=TRUE)
```

7 Solutions to the exercises

- Compare `nrow(dataExt)`, `sum(dataExt$WGTVICCY)`, and `sum(dataExt$SERIES_WEIGHT)`. What does each number represent?

```
nrow(dataExt)
```

```
[1] 9539
```

```
sum(dataExt$WGTVICCY)
```

```
[1] 18201324
```

```
sum(dataExt$SERIES_WEIGHT)
```

```
[1] 20121320
```

`nrow(dataExt)` is the unweighted number of records in our extract. `sum(dataExt$WGTVICCY)` is the estimated number of victimizations when every series record is counted once. `sum(dataExt$SERIES_WEIGHT)` is the estimated number of victimizations after counting up to 10 occurrences of each series crime.

2. Calculate the percentage increase from `WGTVICCY` to `SERIES_WEIGHT`. This is the effect of accounting for series victimizations on the estimated total.

```
100*(sum(dataExt$SERIES_WEIGHT)/sum(dataExt$WGTVICCY)-1)
```

```
[1] 10.54866
```

3. Using `crimeType`, calculate the weighted percentage of all victimizations in each of the three categories: violent crime, personal larceny, and household property crime

```
dataExt |>
  summarize(count=sum(SERIES_WEIGHT), .by=crimeType) |>
  mutate(percent=100*count/sum(count))
```

	crimeType	count	percent
1	property	13582077.9	67.5009293
2	violent	6379364.0	31.7045010
3	personal larceny	159877.9	0.7945696

4. Estimate the number and weighted percentage of motor vehicle thefts that were attempts rather than completed thefts

```
dataExt |>
  filter(V4529 %in% c("(40) Motor veh theft",
                     "(41) At mtr veh theft")) |>
  summarize(count=sum(SERIES_WEIGHT), .by=V4529) |>
  mutate(percent=100*count/sum(count))
```

	V4529	count	percent
1	(41) At mtr veh theft	193058.2	27.12043
2	(40) Motor veh theft	518797.1	72.87957

- Repeat the demographic analysis using marital status. For each marital-status group, calculate the weighted percentage of personal victimizations in every crime category. Which crime types make up a larger percentage among divorced respondents than among married respondents?

```
dataExt |>
  filter(crimeCode <= 23,
         marital %in% c("(1) Married",
                        "(2) Widowed",
                        "(3) Divorced",
                        "(4) Separated",
                        "(5) Never married")) |>
  summarize(count=sum(SERIES_WEIGHT),
            .by=c(V4529, marital)) |>
  pivot_wider(names_from=marital,
              values_from=count,
              values_fill=0) |>
  mutate(across(-V4529, function(x) 100*x/sum(x)),
         difference=`(3) Divorced`-`(1) Married`) |>
  arrange(desc(difference)) |>
  print(n=Inf)
```

A tibble: 22 x 7

V4529	`(1) Married`	`(5) Never married`	`(3) Divorced`	`(4) Separated`
<fct>	<dbl>	<dbl>	<dbl>	<dbl>
1 (11) Ag asl~	3.98	5.94	12.3	6.07
2 (14) Simp a~	5.58	9.09	10.3	23.3
3 (05) Rob w ~	1.01	0.920	3.52	0.720
4 (15) Sex as~	0.116	2.76	1.83	0.526
5 (09) At rob~	0.594	0.557	1.90	0
6 (07) Rob wo~	3.56	2.55	4.57	1.50

7 (06) Rob w ~	0.125	2.29	0.929	0
8 (23) Pocket~	1.22	2.23	1.82	1.19
9 (01) Comple~	1.21	2.53	1.75	1.70
10 (16) Unw se~	0.520	1.01	0.877	0
11 (02) Attemp~	0.818	1.24	1.15	0.246
12 (04) Sex as~	0	0.113	0.148	0
13 (03) Sex as~	0	0.217	0	0
14 (08) At rob~	0	0.747	0	0
15 (21) Purse ~	0.222	0.887	0.188	0
16 (18) Verbal~	0.737	0.625	0.686	0
17 (19) Ver th~	0.492	0.146	0.266	0
18 (10) At rob~	2.87	1.60	0.917	2.84
19 (20) Verbal~	33.1	28.1	30.4	37.0
20 (12) At ag ~	7.16	6.16	4.43	6.74
21 (13) Thr as~	13.4	9.22	9.07	4.28
22 (17) Asl wo~	23.2	21.1	12.9	13.9

i 2 more variables: `(2) Widowed` <dbl>, difference <dbl>

Fisher, Bonnie S., and Rachel L. Gross. 2025. "The Evolution of the Measurement of Rape and Sexual Assault over 50 Years: Milestones, Definitions, Operationalizations, and Classifications." *Journal of Contemporary Criminal Justice* 41 (1): 166–95. <https://doi.org/10.1177/10439862241290352>.